Rain is naturally acidic (pH5), but burning fossil fuels that release polluting **nitrogen oxides and sulfur dioxide** makes it more acidic.

Acids and bases

Chemical opposites, acids and bases react when they are mixed together, neutralizing one another. Bases that are soluble in water are called alkalis. All alkalis are bases, but not all bases are alkalis.

Bases and acids can be weak or strong. Many ingredients in food contain weak acids (vinegar, for instance) or alkalis (eggs), while strong acids and alkalis are used in cleaning products and industrial processes. Strong acids and alkalis break apart entirely when dissolved in water, whereas weak acids and alkalis do not.

Corrosive power

Strong acids and alkalis can cause serious burns to skin. Very strong acids and alkalis can burn through metal, and some can even dissolve glass. While dangerous, their corrosive power can be useful, for instance, for etching glass or cleaning metals.

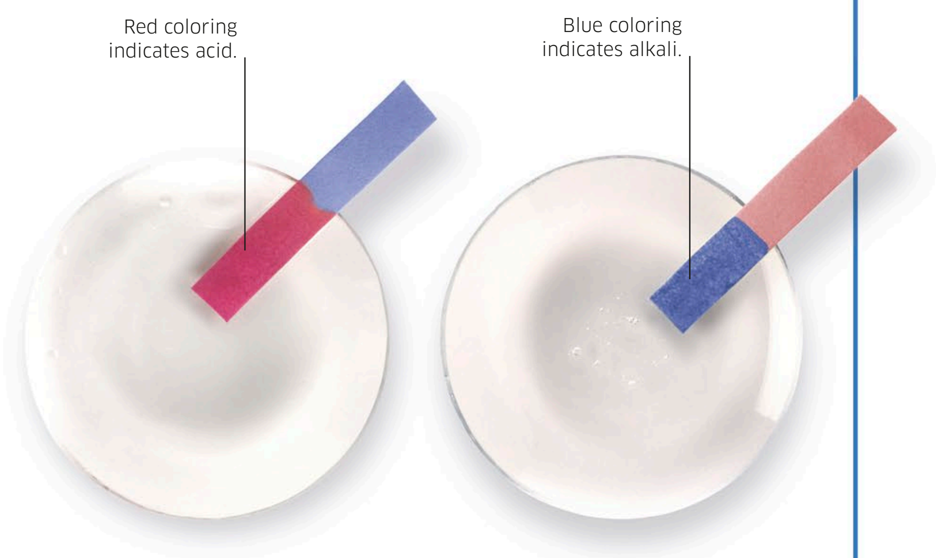
Is it an acid or a base?

The acidity of a substance is measured by its number of hydrogen ions—its "power of hydrogen" or pH. Water, with a pH of 7, is a neutral substance. A substance with a pH lower than 7 is acidic; one with a pH above 7 is alkaline. Each interval on the scale represents a tenfold increase in either alkalinity or acidity. For instance, milk, with a pH of 6, is ten times more acidic than water, which has a pH of 7. Meanwhile, seawater, with a pH of 8, is ten times more alkaline than pure water.

Hydrogen ions (H^+)
determine whether a solution is an acid or an alkali. Acids are H^+ donors while alkalis are H^+ acceptors.

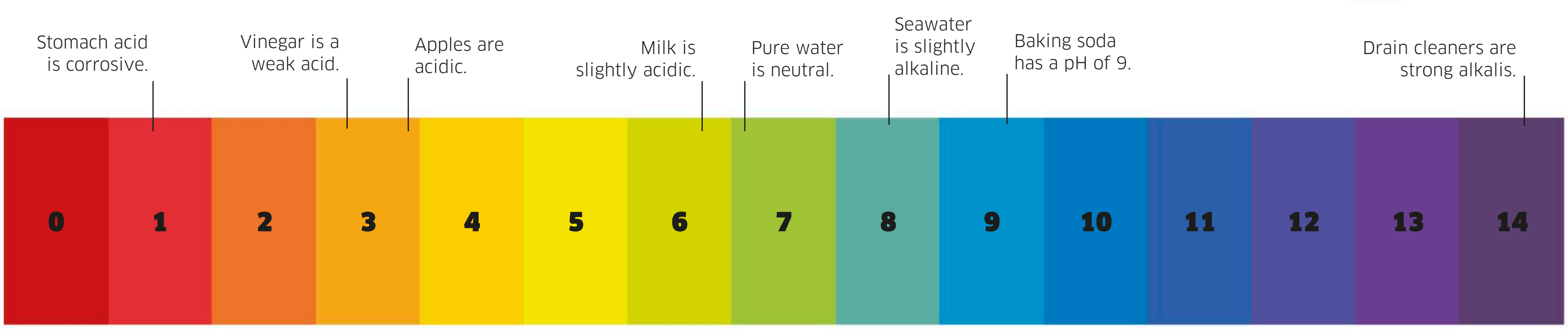
The litmus test

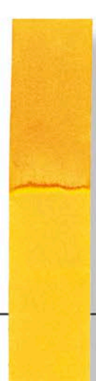
A version of the litmus test has been used for hundreds of years to tell whether a solution is acidic or alkaline. Red litmus paper turns blue when dipped into an alkali. Blue litmus paper turns red when dipped into an acid.



The pH scale

Running from 0 to 14, the pH scale is related to the concentration of hydrogen ions (H^+). A pH of 7 is neutral. A pH of 1 indicates a high concentration of hydrogen ions (acidic). A pH of 14 shows a low concentration (alkaline).





The universal indicator test

Indicator paper contains several different chemicals that react, turning a range of colors in response to different pH values. Dipping indicator paper into an unknown solution reveals its pH.